Course Introduction

Welcome to Python Programming

- Video: Introduction to Programming 4 min
- Video: Why Python? 2 min
- Reading: Visual Studio Code 5 min
- Reading: Installing Python paths (Optional for Windows Users) 5 min
- Reading: Installing Python paths (Optional for Mac users) 5 min
- Reading: Required dependencies 5 min
- Video: Environment check for Windows 2 min
- Video: Environment check for Mac 3 min
- Video: Running code - Command line VS IDE 3 min
- Video: Python syntax, spaces matter 3 min
- Reading: Python syntax cheat sheet
- Reading: Commenting code 5 min
- Video: Variables 6 min
- Video: Basic data types 6 min
- Video: Strings 5 min
- Reading: Basic Data type and Function Cheatsheet 10 min
- Video: Type casting 2 min
- Video: User input, console output 8 min
- Reading: Type casting, a deeper look 10 min
- Programming Assignment: Type casting input 3h
- Practice Quiz: Knowledge check - Welcome to Python Programming 5 questions
- Reading: Additional resources 5 min

Control flow and conditionals

# Basic Data type and Function Cheatsheet

Here's a quick reference for data types in Python.

### Data types

| Data type | Meaning | Example |
|---|---|---|
| string | Text | 'Hello', 'Testing 123' |
| integer | Numbers | -5, 4, 3, 2, 0 |
| float | Decimals | 2.4, 5.2, 1000.00 |

### Flow Control

#### Comparison operators

| Operator | Meaning | Example |
|---|---|---|
| == | Equals | a == b |
| != | Not Equal | a != b |
| < | Less than | a < b |
| > | Greater than | a > b |
| <= | Less than or Equal to | a <= b |
| >= | Greater than or Equal to | a >= b |

### Comments

#### Single-line comments

Placing a # symbol in front of the text you want to be a comment causes Python to ignore everything from that point until the end of the current line.

```
# Single Line comment

```

Run

Reset

```
No Output
```

#### Multi-line comments

Python does not really have a method for multi line comments, so a # symbol can be used on every line.

```
# This is a multiline comment
# which can be used for long comments

```

Run

Reset

```
No Output
```

#### Inline/code comments

The # symbol will cause Python to ignore everything from that point until the end of the current line, so inline comments can be created in this way.

```
x = 1 # assigns value of 1 to x

```

Run

Reset

```
No Output
```

### Built-in Functions

#### print()

This function looks for the default output device, your terminal, and displays the value passed to it.

```
print("Hello")

```

Run

Reset

```
Hello
```

#### input()

This function looks for the default input device, your keyboard, and captures the value. This value can then be assigned or used.

```
print("Where do you live?")
location = input()
print("So you live in " + location)
```

Run

Reset

```
Error on line 2:
    location = input()
EOFError: EOF when reading a line
```

#### len()

This function returns the length or the count of the elements contained within the structure it is applied on. This may be a string, array, list, tuple, dictionary or any sequence.

```
len("Hello")
5
```

Run

Reset

#### str()

This function can be used to convert the provided value into a `String`

```
str(55)
'55'

```

Run

Reset

```
No Output
```

#### int()

This function can be used to convert the provided value into an `int`

```
int('75')
75
```

Run

Reset

```
75
```

#### float()

This function can be used to convert the provided value into a `float`

```
some_int = 10
float(some_int)
10.0

```

Run

Reset

```
10.0
```

### Creating Functions

Functions in Python require a keyword to define them : `def` followed by an identifier (a name) this forms the function signature. The body of the function contains the code to run when the function is called.

```

def say_hello():
    return "Hello there!"

# With parameters
def say_hello(you):
    return "Hello " +  you

```

Run

Reset

✓ Completed    Go to next item

Like    Dislike    Report an issue